

The Tempotron Learning Surface: a Foundations Campaign

Six verifier-gated workstreams on why gradient descent finds solutions in a shattered space
(2026-08-19 – 2026-08-21)

snn-research / tempotron-capacity program

2026-08-21

Abstract

We report a three-day, six-workstream research campaign on the foundations of tempotron capacity, organized around a convex-geometry reframing of the learning problem: the readout $g(w) = \max_t V(t; w)$ is convex piecewise-linear in the weights, negative patterns carve convex (AND) constraint sets, positive patterns carve complements of convex sets (OR), and the version space is a union of convex cells indexed by per-pattern decision-time assignments. Every empirical claim passed an independent numerics-verifier gate; every experiment tested a pre-registered prediction. Headline results: (1) the findable–existence gap is a *search-time* phenomenon — solution cells are certified plentiful exactly where learning fails, the learner never freezes its decision-time assignments at the wall (a new dynamical order parameter), and the findable capacity creeps with training budget, making every findable number budget-conditioned; (2) gradient descent is a *late-settling, margin-biased router*: it strongly prefers high-margin cells (percentile 0.83→0.92 toward the wall) yet under-weights the largest cells relative to volume, and parks near the boundary of its chosen cell; (3) eight standing hypotheses were refuted by their own pre-registered tests, including two of our own theories (the antipodality second axis; the participation-controlled onset); (4) one new unexplained fact emerged — a shared +0.45 capacity lift for line-spectrum (comb) kernels that no registered theory predicts. Total lab spend $\approx$ \$26.

1 Framing and organization

The campaign was organized by a question ledger (`notes/FOUNDATIONS-QUESTION-LEDGER.md`) derived from a first-principles pass over the tempotron: $V(t; w)$ is linear in w at fixed t ; classification thresholds $g(w) = \max_t V(t; w)$, a convex piecewise-linear function whose facets are decision times. A pattern labeled $-$ imposes an AND over $\sim K_{\text{eff}}$ half-spaces (convex); a $+$ pattern imposes an OR (complement of convex). Choosing one decision time per $+$ pattern reduces the problem to a perceptron; the version space is the union of these convex cells, and learning alternates convex margin work within a cell with discrete reassignment events (cell hops), whose gradient is the PSP vector at the argmax (Danskin). Findability is therefore a question about the walk on the assignment structure.

Three literature sweeps (Tavily agents) confirmed the campaign’s three architectural questions were open: constant-across-pattern inputs as a learnable temporal threshold; single-spike timing sensitivity of trained solutions; and per-decision participation (measured exactly once, Gütig & Sompolinsky 2006, never theorized).

Six workstreams ran under a team template (Fable lead; python-pro implementers; independent numerics-verifier owning gates; codex cross-checks on theory), in three gated waves. A binding

scaling protocol required every run to declare which of $\{\alpha, K_{\text{eff}}, N_{\text{dec}}, \text{density}, \text{budget}, \theta/\|w\|\}$ were held fixed versus moved.

2 The wall is search time, not existence

Three independent, verifier-gated measurements converge:

Dynamics at scale (WS-OBS). Logging the per-pattern argmax time each epoch: below the findable wall the assignment settles *late* (last switch at 75–90% of convergence time), and at the wall ($\alpha = 3.0$, unconverged seeds) switching never freezes — de-aliased reassignment rate flat at ≈ 0.32 per logged transition through the full 32k-epoch budget (Fig. 1). Sustained assignment cycling *is* the wall, giving the program its first dynamical order parameter for the findability transition.

Exact ground truth (WS-TOY). At toy scale ($N=16$, $K=6$ bins) with LP-certified exhaustive enumeration of cells: existence is certified in 22/22 instances with median lower bounds of 385–3510 cells at loads where the faithful GD rule fails 25–71% of instances; at $\alpha = 3.0$, 4/5 high-budget instances certify 11–476+ cells while $P_{\text{solve}} = 0/24$ even at 64k epochs (Fig. 2).

Budget creep. The findable wall moves with training budget: $\alpha_f = 2.163/2.309/2.550/2.600$ at 2k/8k/32k/64k epochs (paired flips $p \approx 1.2 \times 10^{-4}$). **Consequence: “findable capacity” is not a constant of the model but a budget-conditioned family $\alpha_f(\text{compute})$; every findable number in the program is a statement at a stated budget.**

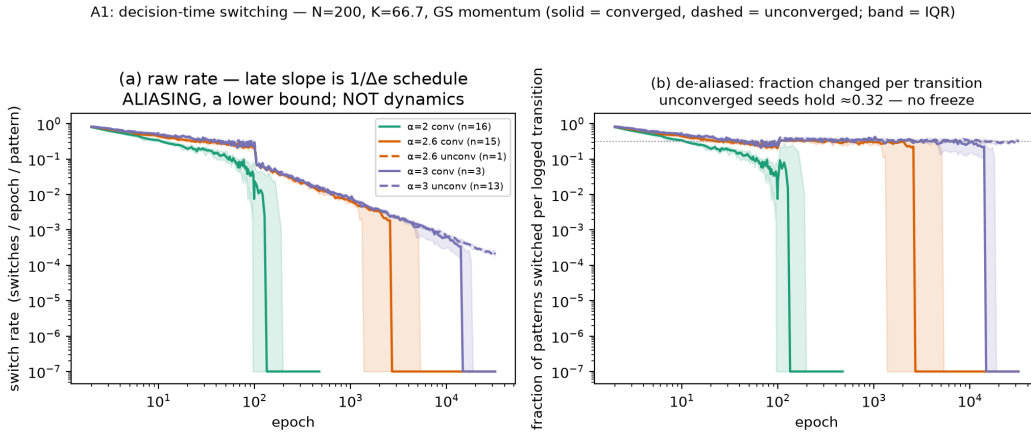


Figure 1: Decision-time switch statistics (WS-OBS, A1). Converged seeds settle late; unconverged seeds at the wall never freeze — the de-aliased switching rate stays flat through the entire budget. The wall is visible as sustained assignment cycling.

3 How gradient descent searches: a measured signature

No imprinting. The initialization’s argmax predicts the final decision time for only 7–12% of patterns (chance 3%; predicted $> 50\%$) — **the greedy lock-in / exploration-failure hypothesis is refuted** (Fig. 3).

Margin-biased cell selection, but not volume sampling. Against exact enumeration of all feasible cells, GD’s chosen cell sits at mean percentile 0.833 (median 0.887) of the max-margin distribution (Wilcoxon $p = 1.6 \times 10^{-11}$, $n = 64$), a bias that *grows* toward the wall ($0.76 \rightarrow 0.92$). Yet a volume-weighted null scores 0.934 — GD (0.748 in the gated 5-instance comparison) is neither

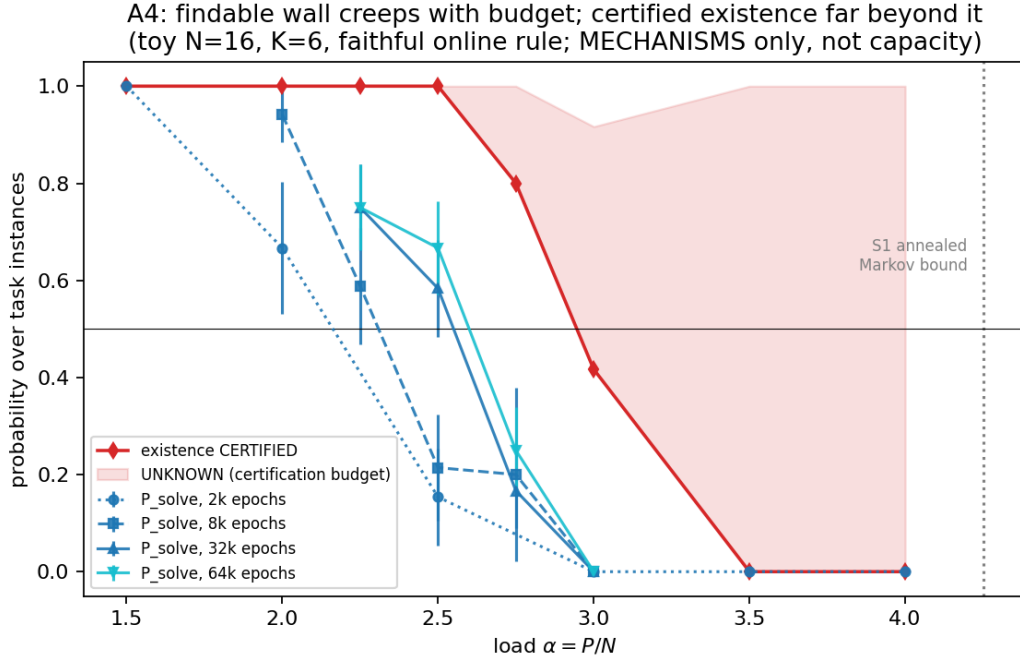


Figure 2: Toy-scale walls vs training budget (WS-TOY, A4). The findable wall creeps monotonically with budget while certified existence stays far above it.

a uniform nor a fair volume sampler; it under-weights the largest cells relative to version-space volume (Fig. 4).

Boundary parking. Within its chosen cell, GD’s margins are $\approx 0.50\times$ version-space-typical (validated micro-scale sampling, 5/5 instances). Synthesis: **GD routes to large, high-margin cells but parks near their boundaries** — the temporal-domain analog of the Baldassi-et-al. algorithmic-bias picture, here measured against exact ground truth.

The stored code rejects superpositions. Convex mixtures of two learned $+$ patterns are winner-take-all in decision time (55% never leave an endpoint bin; median exactly one discontinuous argmax jump), and only 22% of $++$ mixtures remain classified $+$ (Fig. 5).

No kernel-peak placement bias. The pre-registered mechanism “GD parks decisive spikes where $K'(\text{lag}) = 0$, buying jitter robustness” is **refuted with the opposite sign**: trained-vs-null lag mass shows a slight *deficit* at the kernel peak (ratios 0.99–1.00; 14/49 cells exclude 1 from below). GD’s robustness bias lives in cell choice, not spike placement (Fig. 6).

4 Participation and finite size: two laws, decoupled

The per-decision participation law holds at leading order: $N_{\text{dec}} \approx cN/K_{\text{eff}}$ with $c = 3.26\text{--}4.21$ (median 3.43); a τ -free K -move test gives ratio 0.702 (prediction 0.667), excluding N -only scaling; exact proportionality is refuted (slope 0.865 [0.852, 0.878]), with roughly half the drift reproduced by a matched random-weight null (Fig. 7).

But the pre-registered onset decider (\$0.53) **refuted the mechanism we proposed**: an adam-cosine column at $N=350$, $K=100$ ($N_{\text{dec}} \approx 12$) crosses at $\hat{\alpha}_c = 3.363$ [3.282, 3.417] — statistically identical to the $N=350$, $K=66.7$ curve and decisively off the N_{dec} -matched $N=200$ curve (0/20000

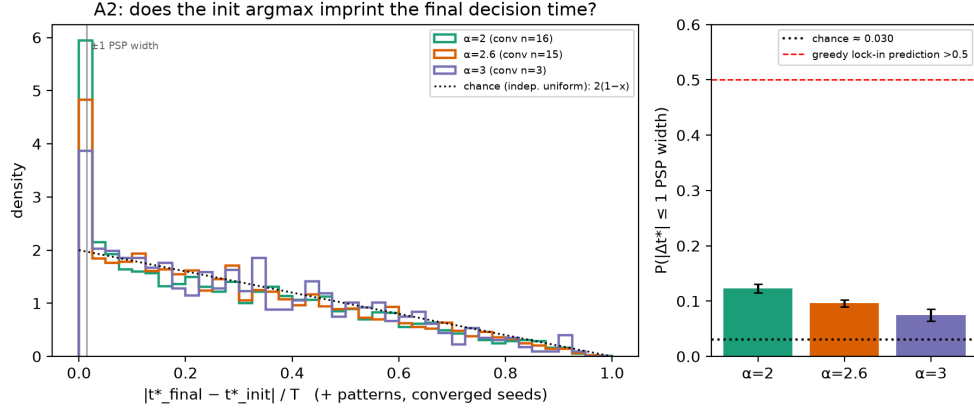


Figure 3: Imprinting test (WS-OBS, A2): the final decision time is essentially not predicted by the initialization's argmax — refuting greedy lock-in.

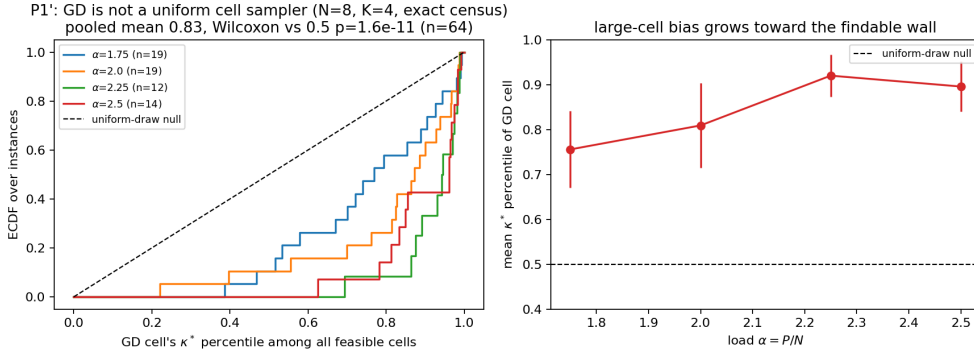


Figure 4: Cell-selection bias (WS-TOY, P1'): GD's cell percentile in the exact max-margin distribution over all feasible cells, vs uniform and volume-weighted nulls.

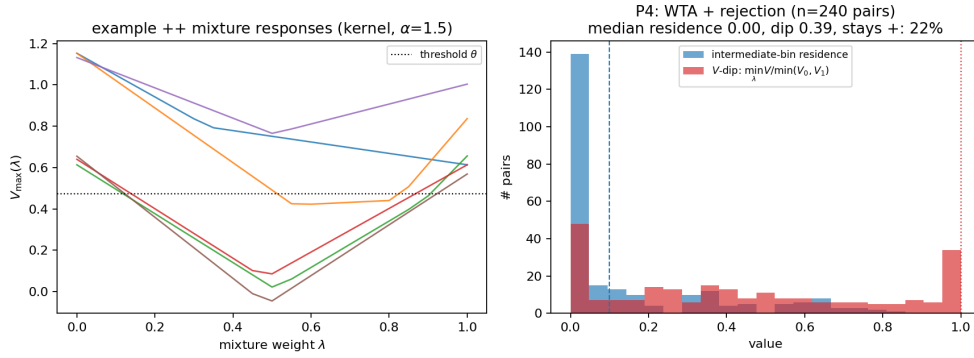


Figure 5: Superposition probe (WS-TOY, C5): mixtures of stored patterns are rejected, not interpolated — winner-take-all decision times with discontinuous jumps.

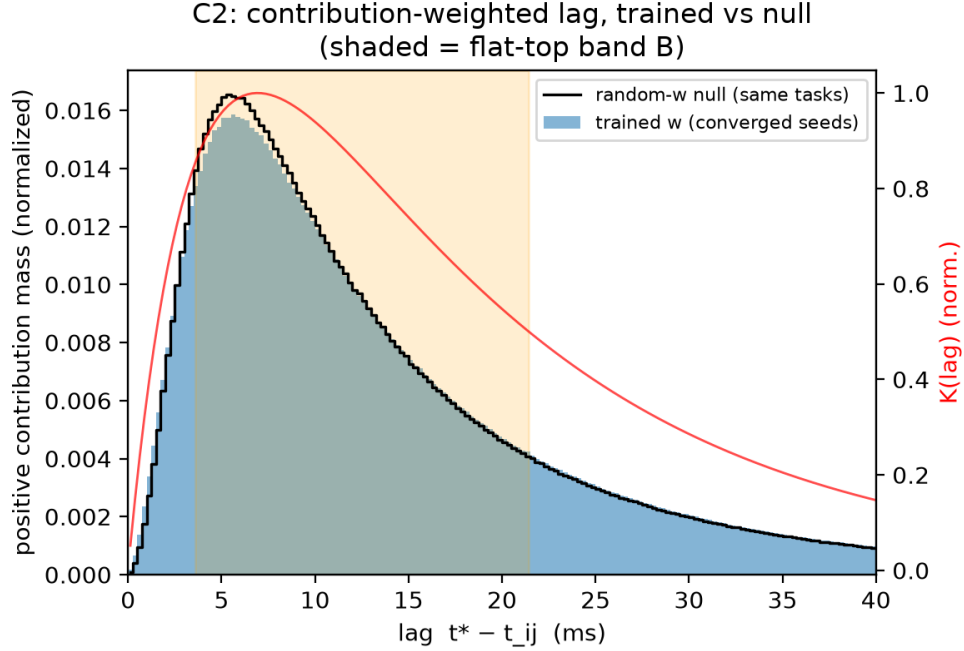


Figure 6: Spike-lag audit (WS-OBS, C2): contribution-weighted lag histograms show no excess mass at the kernel peak ($K'=0$); the flat-minimum-in-time hypothesis is refuted at the $\sim 1\%$ scale.

bootstrap crossings). **The finite-size findable onset follows N , not N_{dec} ; the participation law and the finite- N onset require separate mechanisms** (Fig. 8).

5 Capacity theory: criteria and axes

Learnable temporal threshold (C1). Theory (derivation 13) showed the planned rank- l ladder is a near-null (total span ≤ 0.05 in α_c ; the uniform mode carries the gain), cancelling $\sim \$15$ of planned experiments. The live question became a criterion fight, decided by a threshold-tilt curve at $K=64$ after an exact $K=1$ calibration gate passed at $|\Delta| \leq 0.016$ against the CGMT closed form $\alpha_c = 2/(1 + \tilde{\theta}^2)$ (Fig. 9). **Result: mixed, with the RS sign confirmed** — wing asymmetry $S = +0.93 [+0.76, +1.10]$ and peak $m^* = -0.074 [-0.084, -0.062]$ both carry the RS sign (annealed excluded on both; codex’s on-record prediction that vanilla GD exhibits the RS sign is confirmed), while the curvature $3.04 [2.76, 3.45]$ is $\sim 2.5\times$ super-RS — the pre-registered “neither criterion’s shape” outcome. The two wings fail by different mechanisms (budget-edge slow convergence vs genuinely stuck walls) (Fig. 10).

Label fraction (B1). Theory (derivation 14) refuted our own premise that $f \rightarrow 0$ is convex — it is the negative spherical perceptron (RSB regime) — and predicted a monotone $\alpha_c(f)$ with f -independent $\ln \ln K$ slope. The f -sweep (partial coverage after infrastructure losses) **refutes the routing-only picture** ($\hat{\alpha}(f)$ strongly monotone: $K=64$, $f=1 \rightarrow 3.90$; $f=0 \rightarrow$ findable cliff < 2.2) but delivers *no verdict* on the cluster-vs-RS span — reported as a gap, not stretched. Standout: at $f=0$ the findable cliff sits $\geq \sim 1.95\times$ below the lifted existence estimate (4.30 ± 0.09) — the largest findable–existence gap measured in the program (caveat: learning rate tuned at $f=1/2$) (Fig. 11).

Second axis (B2). Our antipodality proposal $\Gamma = -\min_s \rho(s)$, which retrodicted all 22 archived kernel capacities (RMS 0.117), was put to a pre-registered kill test: an odd-harmonic

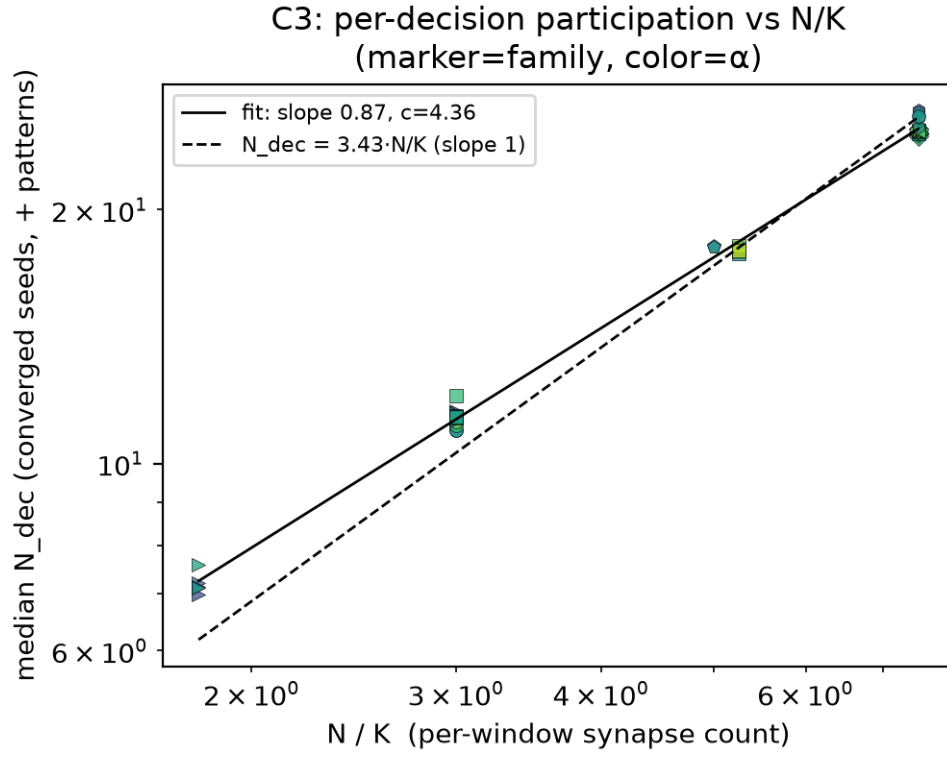


Figure 7: Participation scaling (WS-OBS, C3): N_{dec} vs N/K across the archived grid; leading-order law confirmed, exact proportionality excluded.

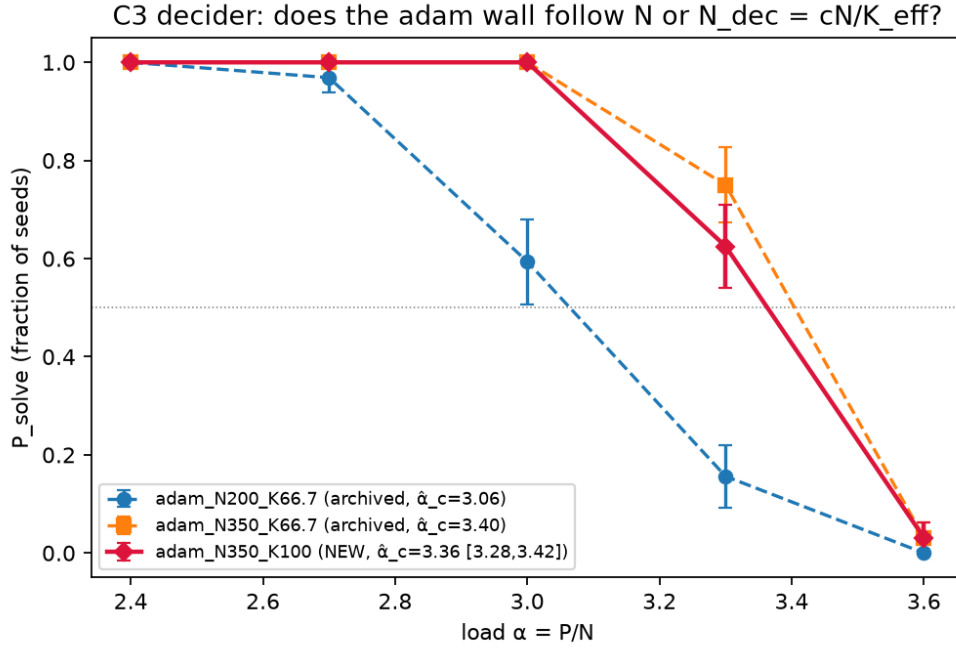


Figure 8: Onset decoder (WS-LAB-KERNELS): the new $N=350/K=100$ crossing lands on the N -matched curve, not the N_{dec} -matched one.

comb ($\Gamma = 0.965$) vs a matched generic comb ($\Gamma = 0.630$) with K_{eff} and spectral shape matched to $\leq 0.25\%$. **The Γ channel is dead:** folded $n=16$ paired $\Delta\hat{\alpha}_c = +0.017 [-0.003, +0.058]$, excluding the predicted $+0.19$ by $3.3\times$ (worst case after nuisances $+0.078$). **New fact: both combs sit $\approx +0.45$ above the same-recipe sinc anchor at matched K_{eff} — a shared line-spectrum lift no registered theory predicts** — now the sharpest open capacity fact in the program (Fig. 12).

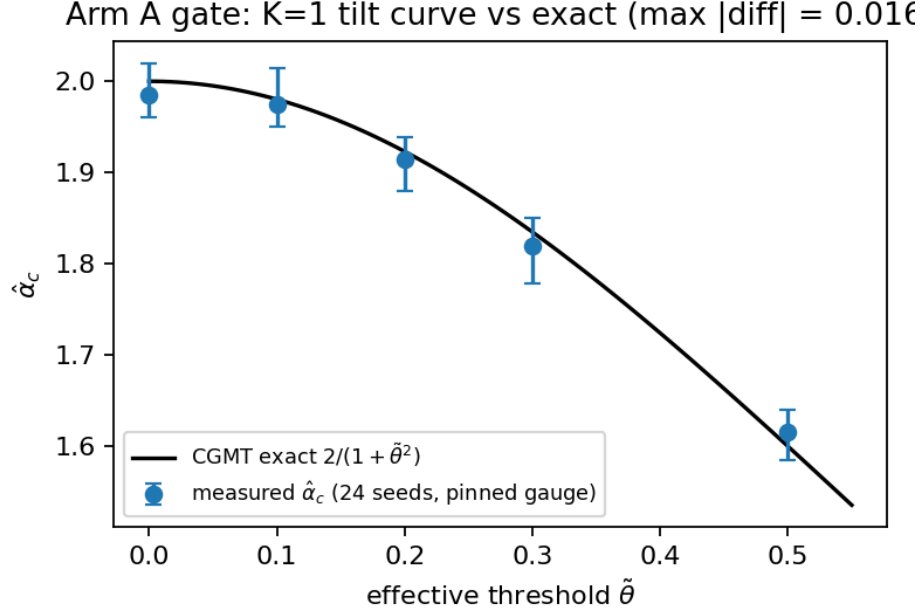


Figure 9: Exact calibration gate (WS-LAB-FIELDS): $K=1$ tilt crossings vs the CGMT closed form; max deviation 0.016.

6 Falsification ledger

Hypotheses killed by their own pre-registered tests during this campaign:

1. Imprinting / greedy lock-in (A2; at-scale and toy).
2. Early assignment freeze (A1; settling is late).
3. Kernel-peak spike placement (K' flat-minimum-in-time) (C2; opposite sign).
4. N_{dec} -controlled finite-size onset (C3 decider; follows N).
5. Antipodality Γ as the second capacity axis (comb; $3.3\times$ exclusion).
6. “ $f \rightarrow 0$ is convex/exactly solvable” (theory; negative perceptron).
7. Homogeneous rank-ladder capacity gain (theory near-null; experiment cancelled).
8. “Cells are rare at the wall” (certified existence; cells plentiful).

Additionally the annealed-cluster criterion is excluded on *sign* by the tilt data, while RS is excluded on *curvature* — both criteria survive only partially.

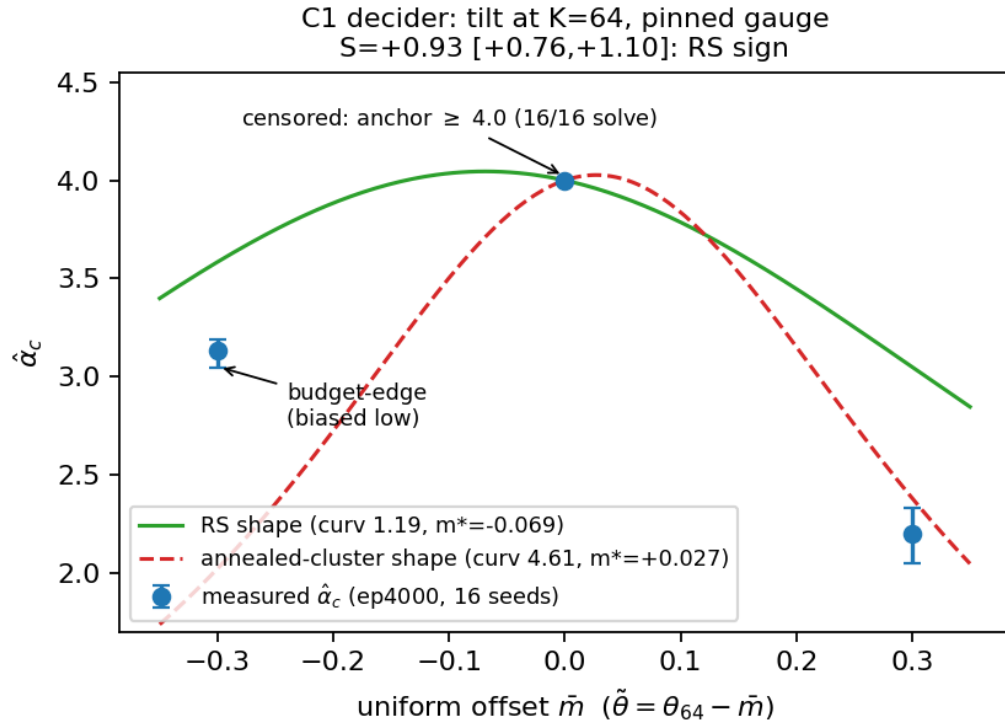


Figure 10: Threshold-tilt curve at $K=64$ (WS-LAB-FIELDS): RS sign on asymmetry and peak location; super-RS curvature — neither criterion's full shape.

B1 decider (partial coverage): f-sweep, pinned gauge
 monotone rise confirmed; $f=0$ findability collapse $\geq 2x$ below exist

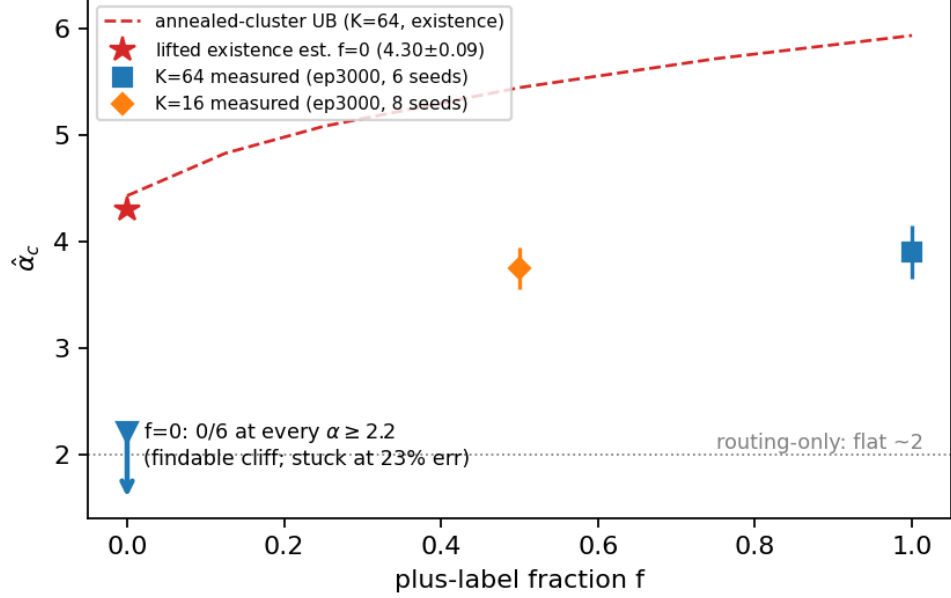


Figure 11: Label-fraction sweep (WS-LAB-FIELDS, partial): findable capacity is strongly monotone in the $+$ fraction; the $f=0$ findable cliff sits far below existence.

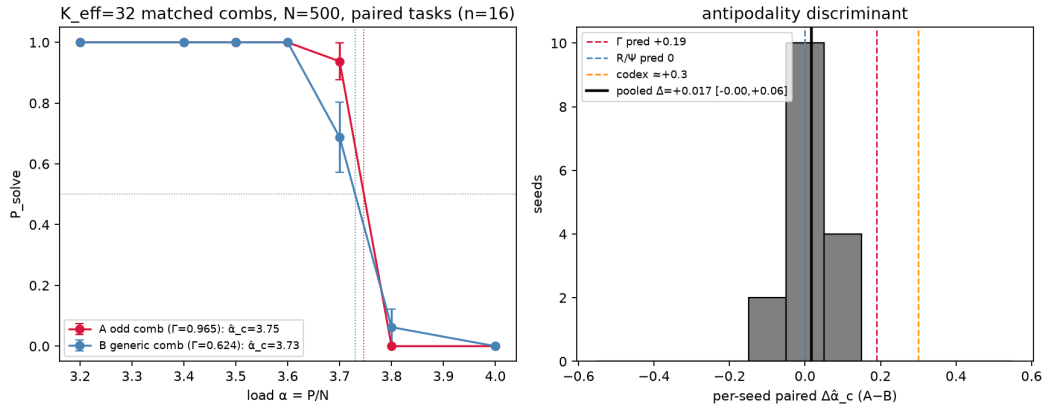


Figure 12: Odd-harmonic comb decider (WS-LAB-KERNELS): the pre-registered Γ kill branch fires; both combs share a large unexplained lift over the sinc anchor.

7 Instrumentation findings

The comb verification exposed a latent harness fault: in the v13 kernel harness, `seeds=` is a label that never enters task generation (tasks derive from `cell_seed`), so sharded “seeds 8–15” jobs silently re-ran the same 8 tasks. The comb result was remedied with verified-distinct top-up tasks; a standing audit item covers previously published sharded 16-seed kernel-campaign results (including the sinc anchor), whose effective n may be 8. The v16 fields harness was verified immune. Operationally, two external VM die-offs (11 jobs) motivated per- α sharding with sub-cell incremental flushes, which bounded losses in the final wave.

8 Conclusions and the road ahead

The campaign inverts the program’s central question. “Why does gradient descent find solutions in a shattered space?” presumed the space was the obstacle. It is not: solutions are certified plentiful, the per-cell geometry is benign, and GD carries a measurable, useful bias toward robust cells. What limits learning is *routing time* — sustained assignment cycling with a measured order parameter. The sharp next questions are: what sets the cycling time-scale and its divergence with α ; whether the margin bias is cause or symptom of slow routing; the origin of the unexplained line-spectrum capacity lift; and completing the two criterion adjudications (costed top-ups: $\approx \$2.2$ tilt, $\approx \$2.5$ –3 f-sweep).

Provenance. All claims verifier-gated; state, scripts, gates, and spend ledgers under `experiments/foundations_*` derivations 13–15 under `lit/tempotron-reductions/derivations/`, question ledger and synthesis under `notes/`. Total lab spend $\approx \$26$ of a \$60 authorization; two incidents (harness crash, user VM stops) cost $\approx \$3$ and no data that was not resubmitted.